

I. (YUCT):

К Ъ

К Ъ ,

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YUCT: <https://yuct.org/>

YPSDC: <https://ypsdc.com/>

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.2.1.

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К НБ (YUCT). , , . , КНБ : , . - ( ) .  
 ,  $\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}$   $\beta = 0.67$ , 50 ( L) : ( , GWTC-4.0), ( , 26r041  
 ), ( ), (873 ), (68 ) (- , UCD). , (UCD), H,  $\alpha_D, \beta_I, \gamma_R$ , .  
 : (1) ; (2) ; (3) ( , ); (4) / . YUCT К НБ, , , .

: YUCT, , , , , , , UCD, ,

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1 : К Ъ

К Ъ, , . , ( ), .  
:

1. () : , .

2. / : .

3. : .

4. : .

: , . YUCT , , . К Ъ, , (  $K_{\text{eff}} \gg 1$  ).  
YUCT , .  $\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}$   $\beta = 0.67$  50 ( L ): (873 , C) (218 , G),  
(26r041 , P) (68 , E). , (UCD), H,  $\alpha_D, \beta_I, \gamma_R$ , . ( , , ) ( H). , , .

## 2 YUCT:

### 2.1 : YPSDC

К Ъ, , : , ()? К Ъ, , ? [6]. YUCT , , , .

#### 2.1.1 YPSDC

(YPSDC). , К Ъ, , К Ъ . entry, .  
. ; ( $D_2$ ). ( К Ъ 650 ) , - ( entry К Ъ ). К Ъ- YPSDC.  
, К Ъ-. ( 650 , ) , entry К Ъ. entry , ( $K_{\text{eff}} \gg 1$ ), . К Ъ .  
К Ъ К Ъ; , .

#### 2.1.2 :

. B- ( К Ъ ), . B- entry. К Ъ, . ( ) entry ( B), , .  
YPSDC : , [34].  
, , , . entry . К Ъ , , .

## 2.2

. YUCT : , ( ).  
:

$$\Psi(t) = D(t) + I(t) \times R(t)$$

:

- $D(t)$  ( , , );
- $I(t)$  ( , );
- $R(t)$  ( ).

,  $R(t)$  , .  $K_{\text{eff}} = \alpha_D \cdot \beta_I \cdot \gamma_R$ , H. (Zhao et al. 2025, Kumar et al. 2024, Lutz et al. 2017),  $K_{\text{eff}} = 0.05$  3.6 ,  $K_{\text{crit}} \approx 8.5$  ( H, 3). , , .

## 2.3

КНБ КНБ , , .

$$:$$

$$(\text{КНБ}) , .$$

, КНБ, , :  $(, ) K_{\text{eff}}^{-\beta} \cdot K_{\text{eff}}$ , ;  $K_{\text{eff}}$ , .

## 2.4

КНБ , , , :  $K_{\text{eff}} \rightarrow \infty$ , КНБ. КНБ,  $K_{\text{crit}} \approx 8.5$ , (Т).

## 2.5

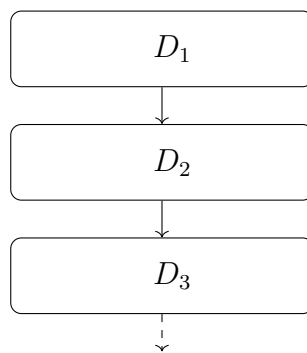
YUCT . КНБ,  $\mathbf{C}_{\text{pers}}$ .  
 $120 \text{ (A)} \quad \kappa_{sr} \quad \gamma_R^{(i)} :$

- $(, , , \text{КНБ} \{K_{\text{eff}}, R, \varepsilon\})$ ;
- , ;
- ,  $(\Gamma , )$ .

, ,  $D_{\text{meta}}$ ,  $\mathbf{C}_{\text{pers}}$ . , КНБ,  $L$  .  
 , КНБ YUCT ,  $K_{\text{eff}}$  . , КНБ,  $L$  .

## 3 :

1 ( ). YUCT :



. 1: , .

### 3.1 1. $(D_1)$

, . КНБ , -, КНБ, КНБ. :  $\mu \approx 68 \quad \mu \propto K_{\text{repair}}^{-0.67} (E, . 2)$ , , .

### 3.2 2. / ( $D_2$ )

(, ) . КНБ entry. , - ( H). ,  $r$  ,  $1.81 K_{\text{eff}}$ ,  $1.81^{-0.67} \approx 0.65$ , ( D).

### 3.3 3. / ( $D_3$ )

, , , . ( ) YUCT КНБ ( $\kappa$ ) КНБ ( $\Delta D$ ). : ( )  $\gamma \approx 0.38$ ,  $\beta\gamma = 0.20$   
 $\beta = 0.67$  ( T).  $K_{\text{eff}} \approx 10^2$   $K_{\text{eff}} \approx 10^{10}$  ( T).  
 ( 1) , , , , , .

## 4

### 4.1 $\beta = 0.67$

$\beta = 0.67$  ( UCD), .  $1/f$ -,  $\sigma$  :  $\gamma_R \propto |\log_{10}(\sigma)|$  ( H). ,  $\sigma \propto K_{\text{eff}}^{-\beta}$  , , .

### 4.2

, . YUCT  $K_{\text{eff}} \approx K_{\text{crit}}$  . ( , ); , .  $K_{\text{crit}}$ , . H ,  $K_{\text{eff}}$   
 1.01.5, 3.6, .

## 5 :

### 5.1

, , . YUCT, ( $K_{\text{eff}}$ ) . , КНБ КНБ, .

### 5.2

, :

- $D_1$  ( ) :

- ( , )  $K_{\text{eff}}$  .
- ( ) , .
- ( , )  $R$ .
- ( , ) entry.

- $D_1$  :

- (КНБ) , .
- ( = /)  $D_3$ .
- $D_2$   $D_3$  .

, , .



### 5.3 :

:

$$\rightarrow D_1 ( ) \rightarrow D_2 ( ) \rightarrow D_3 ( ) \rightarrow$$

: :

- $D_1$ :  $\rightarrow ( )$
- $D_2$ :  $\rightarrow (/) \rightarrow ( )$
- $D_3$ :  $(\mathbb{K} \mathbb{H}) + (\mathbb{K} \text{ , } \mathbb{H})$
- :

### 5.4 $K_{\text{eff}}$

:

$$\tau_{\text{perception}} \propto \frac{1}{K_{\text{eff}}}$$

$$K_{\text{eff}} ( 10^3 10^4 ) \text{ , , . : } ( ) ( ) \text{ .}$$

### 5.5 ,

:

- :
  - $D_3$ ,  $K_{\text{eff}}$  (Lutz et al. 2017).
  - $D_3$  .
- :
  - $( )$  entry  $( ) K_{\text{eff}}$  .
  - .
- :
  - entry  $D_1/D_2$ , , .
  - $(K_{\text{eff}})$ .
- /:
  - $D_3 (/) D_1/D_2$  .
  - ,  $K_{\text{eff}}$  .

## 6 : YUCT

,  $\mathbb{K} \mathbb{H}$  . YUCT ,  $( )$  .

## 6.1

, . YUCT , ,  $K_{\text{eff}}, R, \varepsilon$ . , :

1.  $D$ , , .
2.  $R$ , .
3.  $\{K_{\text{eff}}, R, \varepsilon\}$ , .

## 6.2

YUCT :

| 1: YUCT. |                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------|
| YUCT     |                                                                                                          |
| $D$      | $\alpha_{\text{D}}$ ( , , , ).                                                                           |
|          | $R$ , , , ( ) .                                                                                          |
|          | $K_{\text{eff}} = \alpha_{\text{D}} \cdot \beta_{\text{I}} \cdot \gamma_{\text{R}}$ . $K_{\text{eff}}$ . |
|          | $\beta_{\text{I}} = H_{\text{int}}/H_{\text{ext}}$ ( $H_{\text{int}}$ ), $\beta_{\text{I}}$ , .          |
|          | $\gamma_{\text{R}} = \tau_{\text{coh}}/\tau_{\text{decay}}$ , .                                          |

## 6.3

,  $H$  ( ??), :

- / :  $K_{\text{eff}} > 1.2$   $R \rightarrow S_{\text{odd}} = 1.2$ . , , . , .
- / :  $0.8 < K_{\text{eff}} < 1.0$   $R \rightarrow S_{\text{even}} = 0.8$ . . ( , ) , .
- / : ,  $K_{\text{eff}} \sim 1.1$ ,  $R$  . , , .

## 6.4

YUCT ( ). :

1.  $K_{\text{eff}}$  (  $H_{\text{int}}$  ) (  $R$  ).
2. YUCT ( , , , ) .
3. , entry ( ) , , , , , , .

$\kappa \mathbb{H}$ , -, , , .

## 6.5

, :

1. : , , .  $D_1, D_2, D_3$ .
2. :  $D_1 \sim D_2$  .  $\kappa \mathbb{H}$ , ( , ,  $D_1$  ).
3. YUCT: .  $\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}$  , , , .

## 6.6 : ?

, :

- : ( entry) , .
- : entry ( $D_2$ ) ( $D_3$ ).
- :  $K_{\text{eff}}$  .
- : entry .

, , К-НБ.

## 7

, YUCT. , YUCT , , .

| 2: / YUCT.          |                                                                               |                                               |
|---------------------|-------------------------------------------------------------------------------|-----------------------------------------------|
|                     | YUCT                                                                          | YUCT                                          |
| ( $\Phi$ ) (Tononi) | ( $\Phi$ )                                                                    | ; $K_{\text{eff}}$ .                          |
| / (Baars, Dehaene)  | ( $R$ )                                                                       | ; К НБ .                                      |
| / (Friston, Seth)   | YPSDC                                                                         | ( ) ( ) ( $\varepsilon$ ).                    |
| / (Damasio)         | $I_{\text{int}}$                                                              | ; КНБ .                                       |
| (Barrett)           |                                                                               | $\{K_{\text{eff}}, R, \varepsilon\}$ , .      |
| / (Fleming)         |                                                                               | ; , $K_{\text{eff}} > K_{\text{conscious}}$ . |
| (Cowan)             | $\alpha_{\text{D}} ( )$<br>$\tau_{\text{update}}^{-1}$<br>$V(K_{\text{eff}})$ | .<br>, $D$ .<br>, .                           |

$$\text{UCD ( H)} = \alpha_{\text{D}}, \beta_{\text{I}}, \gamma_{\text{R}}, \quad \varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta} \quad \beta = 0.67. \quad .$$

## 8 : UCD

( , ) . , , ? YUCT UCD : , ; , .  $\alpha_{\text{D}}, \beta_{\text{I}}, \gamma_{\text{R}}$ , , .

### 8.1 UCD

UCD, ( H), :

- $\alpha_{\text{D}} ( )$ : , .
- $\beta_{\text{I}} ( )$ : ( , , -) .
- $\gamma_{\text{R}} ( )$ : ( , ) .

## 8.2

UCD (H) . , , .

| 3: UCD . |                |                   |                   |                                   |
|----------|----------------|-------------------|-------------------|-----------------------------------|
|          | $\alpha_D$ ( ) | $\beta_I$ ( )     | $\gamma_R$ ( )    |                                   |
|          | $10^2 10^3$    | 0.52.0            | $10^{-2} 10^{-1}$ | : , , , , .                       |
|          | $10^2 10^3$    | 0.10.5            | $10^{-1} 1.0$     | $\gamma_R$ КНБ- ( , , ).          |
|          | $10^2 10^3$    | 0.51.0            | $10^{-2} 10^{-1}$ | , : ( ), /.                       |
| /        | $10^1 10^2$    | 0.050.1           | $10^{-2}$         | : ( ), ( ), ( ). , .              |
|          | $10^2$         | 0.10.2            | $10^{-2} 10^{-1}$ | $(\alpha_D, \beta_I)$ . , , ; / . |
| ( )      | $10^1 10^2$    | $10^{-3} 10^{-2}$ | $10^{-3} 10^{-2}$ | : ( ), ; .                        |

## 8.3 YUCT

YUCT :

1. /: ( , , , ).  $\gamma_R \beta_I$  .
  2. : ( , ).  $\alpha_D$ .
  3. : YUCT , ,  $K_{\text{eff}}$  ( , , ),  $\{K_{\text{eff}}, R, \varepsilon\}$ , .
  4. : , , ( , , ) ( , ). YUCT ,  $\alpha_D \beta_I$  .
- , , YUCT , , .

## 9

YUCT , , . YUCT , .

1. (IIT): , , ,  $\Phi$ . IIT: К, Ё. YUCT  $(\Phi)$ . ,  $\Phi$  IIT , YUCT :  $\Phi$  ( $K_{\text{eff}} \gg 1$ ).
2. (GWT): , GWT , К Ё. , К Ё . К Ё. YUCT КНБ ( $R$ ),  $R$  , , .
3. : (FEP), ( , ) , . К Ё. YPSDC, YUCT. КНБ КНБ, КНБ . , YUCT (D+IЧR) , FEP - .
4. : . , (К Ё) . , , . YUCT К Ё : ( ) , ( )  $\{K_{\text{eff}}, R, \varepsilon\}$ .

## 10 : YUCT

, , . (YUCT) , , , .

### 10.1 (PCM):

, , , : 3D-, , , [35]. .

YUCT

- . К Ъ PCM YUCT.  $(\kappa)$ ; , entry  $(D_2/D_3)$ . , PCM-  $-K_{\text{eff}}$  .
- . 3D-  $D_2$ , . К Ъ .
- YPSDC. К Ъ , К Ъ YPSDC.  $K_{\text{eff}} \gg 1$ , entry , , К Ъ . К Ъ ( , ) , entry YPSDC.
- . К Ъ PCM  $(D_{\text{meta}})$ . , ,  $K_{\text{eff}} K_{\text{crit}} \approx 8.5$ .

## 10.2 (IIT): $\Phi K_{\text{eff}}$

, , ,  $\Phi()$ . , , [7]. IIT : , , , .

YUCT

- $\Phi K_{\text{eff}}$ . YUCT,  $\Phi$  . ,  $\Phi \propto K_{\text{eff}} \cdot I_{\text{mutual}}$ ,  $I_{\text{mutual}}$  .  $K_{\text{eff}} \gg 1$  , .
- IIT YPSDC. IIT YPSDC: ( ), ( entry), ( ), ( ) ( entry ).
- . IIT  $\Phi$  . YUCT :  $K_{\text{eff}}$  PCI TMS-, -, К Ъ.

## 10.3 (GWT): К Ъ

(GWT), (GNWT), , К Ъ. , (К Ъ) [8, 9].

YUCT

- К Ъ  $(R)$ . GWT/GNWT К Ъ -  $(R)$  YUCT. К Ъ  $(\kappa)$  , entry , , entry .
- . К Ъ , ,  $-K_{\text{eff}}$  . entry , , .
- . , , GWT [36]. YUCT  $K_{\text{eff}}$ ,  $\varepsilon$  .

## 10.4 (HOT): -

(HOT), , ( ) [40]. , [41].

YUCT

- $(D_{\text{meta}})$ . HOT YUCT  $(D_{\text{meta}})$ . entry, . ( , ) ,  $D_{\text{meta}}$ .
- . ,  $(D_{\text{meta}})$  , [36]. YUCT ,  $D_{\text{meta}}$  , .
- $D_{\text{meta}}$ . HOT , .

## 10.5 (PP) : YPSDC

, , , ( ) [10, 37]. [38].

YUCT

- YPSDC . PP YPSDC.  $\kappa$   $(\kappa)$ , .  $\kappa$   $(D)$ , .  $\kappa$   $(\varepsilon)$ , .  $(R)$ , .
- , . , PP [39]. YUCT  $K_{\text{eff}} ( \alpha_D, R) \varepsilon$ .
- . YUCT :  $F_{\text{YUCT}} = -\log(K_{\text{eff}}) + \lambda \|\nabla_D I\|^2 / R$ .

## 10.6 (Damasio):

, , , , [12]. , , , .

YUCT

- .  $(\kappa)$ ,  $(D_1 D_2)$ . , (/) , .
- .  $\kappa$   $( )$  , [12]. YUCT,  $D_1 D_2$  ,  $(D_{\text{meta}})$ , , , .
- . ,  $-K_{\text{eff}}$  , , .  $\kappa$   $( )$  , YPSDC , .

## 10.7 (Barrett):

$\kappa$   $( )$  , ,  $( )$  [13].

YUCT

- $\{K_{\text{eff}}, R, \varepsilon\}$ . YUCT . ,  $\kappa$   $(D_3)$  entry  $(, \kappa)$ .  $\kappa$   $( )$   $(D_1/D_2)$ ,  $(I_{\text{int}})$ . , .
  - $\alpha_D$ .  $\kappa$   $( )$  [42]  $\alpha_D$  ,  $(D_3)$  .
  - -  $D_3$ . . YUCT  $(D_3)$ , , .
  - . ,  $\kappa$   $( )$  [43], YPSDC : entry , , .
- , YUCT . , , .

## 11

### 11.1

YUCT , .  $\kappa$   $( )$  . , , , .  $\alpha_D, \beta_I, \gamma_R$  , .

### 11.2

. , , , .  $K_{\text{eff}} ( T) , , . \kappa$  :  $K_{\text{eff}}$  , .

### 11.3

YUCT :  $K_{\text{eff}}$  :

- $(, )$   $K_{\text{eff}} \sim 10^2$  ( O).
  - $\alpha_{\text{D}} \sim 10^5$ ,  $\beta_{\text{I}} \sim 10^{-6}$ ,  $K_{\text{eff}}$ ; ,  $\beta_{\text{I}}$  ( H).
  - $K_{\text{eff}} \gg 1$ ,  $(, , )$ .
- $\kappa_{\text{H}}$ , ,  $K_{\text{eff}}$ . UCD :  $K_{\text{eff}} - K_{\text{crit}} \approx 8.5$ ,  $(\alpha_{\text{D}}, \beta_{\text{I}}, \gamma_{\text{R}})$  , .

### 11.4

$(D)$  , :

- :  $M_D$   $(, -)$ . 19- YUCT ( A) .
- :  $dD/dt$ , , . YUCT  $\Psi_{MN}$ .
- : , 120 ( A).

### 11.5 /

:  $K_{\text{eff}} \rightarrow 0$  .  $\kappa_{\text{H}}$ , , , .  $\Lambda \approx 1/L_0^2$ ,  $L_0 \sim R_H$  ( M). , , , .

## 12 :

: ? YUCT :

### 12.1

$$\mathcal{L}_{\text{dictionaries}} = \mathcal{L}_{\text{fundamental}} + \mathcal{L}_{\text{emergent}} + \mathcal{L}_{\text{creative}}$$

#### 12.1.1 1. :

:

$$\Psi_{\text{initial}} = \sum_{n=1}^N \alpha_n |\chi_n^{\text{fund}}\rangle$$

, , .

#### 12.1.2 2. :

:

$$\frac{d\chi_{\text{new}}}{dt} = \mu \cdot \nabla_{\text{old}} S_{\text{info}} \times \text{Innovation}(t)$$

: , , , . : entry  $K_{\text{eff}}^{-\beta}$  .

#### 12.1.3 3. :

- :

$$\mathcal{L}_{\text{creation}} = \lambda_{\text{create}} \cdot \exp[-\beta \cdot D(\chi_{\text{new}} || \chi_{\text{fund}})] \cdot K_{\text{eff}}^{\text{creative}}$$

$D(\chi_{\text{new}} || \chi_{\text{fund}})$  ,  $\kappa_{\text{H}}$  .





14.2 1. 2 3:

/:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \gamma \left( L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\} \right)$$

$$: K_{\text{eff}}^{(2 \rightarrow 3)} = \frac{\tau_{\text{coh}}}{\tau_{\text{dec}}} > K_{\text{crit}} \approx 1. \quad K_{\text{eff}} \quad ; \quad 1, \quad (, ).$$

14.3 2. 3 4:

$K_{\text{eff}}$ :

$$\dot{\theta}_i = \omega_i + \frac{K_{\text{eff}}}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i)$$

$$: K_{\text{eff}}^{(3 \rightarrow 4)} > K_c = \frac{2}{\pi g(0)}.$$

14.4 3. 4 5:

$K_{\text{eff}} \quad \Phi$ :

$$\Phi = K_{\text{eff}} \cdot I_{\text{mutual}}$$

$$: \Phi > \Phi_{\text{crit}} \approx 20 - 30 \quad ( ). \quad \text{IIT} \quad , \quad \Phi, \quad \Phi \quad K_{\text{eff}}.$$

14.5 :

:

$$V(\psi) = \frac{\alpha}{2}(K_c - K_{\text{eff}})\psi^2 + \frac{\beta}{4}\psi^4$$

$$K_{\text{eff}} > K_c, \quad \psi. \quad \beta \quad \beta = 0.67 \quad ( \text{ L} ).$$

14.6 :  $K_{\text{eff}}$

$K_{\text{eff-}}$  :

$$\frac{dK_{\text{eff}}}{dt} = \alpha K_{\text{eff}}(1 - K_{\text{eff}}/K_{\text{eff,max}}) - \beta \nabla^2 K_{\text{eff}} + \gamma \xi(t)$$

$$. \quad K_{\text{eff}} \quad (K_{\text{eff}} \ll 1) \quad , , , , . \quad : \quad K_{\text{eff}}, \quad , \quad .$$

15 YUCT

$$, \quad , \quad \text{YUCT} \quad \text{к н} \quad . \quad \varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta} \quad \beta = 2/3. \quad , \quad .$$

15.1 1:

: , ,  $K_{\text{eff}}$  ,  $K_{\text{eff}} \approx 8.5$ .

: YUCT  $K_{\text{eff}} = K_{\text{crit}}$ .

:  $K_{\text{eff}}$  ( , ) , ( , ) . YUCT , , , .

## 15.2 2:

:  $(, , )$  , , .  $q = (3/2)^{1/3} \approx 1.14$ .  
:  $\{K_{\text{eff}}, R, \varepsilon\}$ ,  $q$ .  $R$ , .  
: ,  $(, )$ .  $\approx 1.14$ , .  $(, -, )$ .

## 15.3 3:

: - -  $(, \rightarrow )$  КНБ,  $(\rightarrow )$ .  
: -  $(, )$   $K_{\text{eff}}$   $R$ .  $(K_{\text{eff}} \approx 1.0)$  , . ,  $K_{\text{eff}}$  - , .  
:  $\tau$   $(, , )$  .  $\tau_{\text{down}}$  ,  $\tau_{\text{up}}$  . YUCT  $\tau_{\text{up}}/\tau_{\text{down}} > 1.5$ .

## 15.4 4:

: , , , , ,  $K_{\text{eff}}, R$   $\varepsilon$ .  
:  $K_{\text{eff}} < 0.8$   $R$ , . ,  $K_{\text{eff}}$   $\varepsilon$ .  
: YUCT . YUCT , ,  $\{K_{\text{eff}}, R, \varepsilon\}$ . ,  $( )$  , , .

## 15.5 5:

: ,  $(, , , )$ , , .  
:  $(, )$ , .  $\alpha_D ( )$   $K_{\text{eff}} R$  .  
: YUCT  $(, , , )$  ,  $(, , )$ . YUCT , , ,  $(, \text{КНБ } 0.8 < K_{\text{eff}} < 1.0)$   
. , .

## 15.6 6: $\beta_I$

:  $(\alpha_D \gg 1)$ ,  $(\beta_I \ll 1)$ . YUCT, , .  
:  $\beta_I = H_{\text{int}}/H_{\text{ext}}$  .  $\beta_I \rightarrow 0$  ; , .  $\alpha_D$ ,  $\beta_I$ .  
: , КНБ . YUCT ,  $\beta_I$  ,  $(\beta_I \ll 10^{-2})$ . ,  $(, , 2 \ 3)$ , . , .

## 15.7 YUCT:

(UCD) , . YUCT  $(N_{idx})$  .

1 ( YUCT).  $N_{idx} \in [0, 1]$  . UCD:

$$N_{idx} = \Phi \left( \frac{\alpha_D}{\alpha_D^{\text{human}}}, \frac{\beta_I}{\beta_I^{\text{human}}}, \frac{\gamma_R}{\gamma_R^{\text{human}}} \right)$$

$\Phi$  ,  $\Phi(1, 1, 1) = 1$ .

, , . ,  $(\alpha_D)$ , , ,  $(\beta_I)$   $(\gamma_R)$ . КНБ ,  $(N_{idx} \ll 0.1)$ . , YUCT , .

## 16 $(L_c)$ :

YUCT UCD , .  $(L_c)$  , , . КНБ КНБ .

### 16.1 $L_c$

YUCT,  $[0, 1]$  :

$$L_c = \sigma \left( w_1 \cdot \frac{K_{\text{eff}}}{K_{\text{opt}}} + w_2 \cdot \frac{\beta_I}{\beta_{\text{opt}}} + w_3 \cdot \frac{\gamma_R}{\gamma_{\text{opt}}} - w_4 \cdot \frac{\varepsilon}{\varepsilon_{\text{base}}} \right) \quad (1)$$

:

- $K_{\text{eff}} = \alpha_D \cdot \beta_I \cdot \gamma_R$  .
- $\beta_I$  , .
- $\gamma_R$  , .
- $\varepsilon$  , КИ, .
- $K_{\text{opt}}, \beta_{\text{opt}}, \gamma_{\text{opt}}$  , (, КИ).
- $\sigma(\cdot)$  ,  $L_c \in [0, 1]$ .
- $w_i$  ,  $\varepsilon$ .

### 16.2 $L_c$

UCD  $L_c$  :

| 4: $(L_c)$ . |                  |      |                                          |
|--------------|------------------|------|------------------------------------------|
| $L_c$        | $K_{\text{eff}}$ |      | YUCT                                     |
| 0.00 – 0.20  | < 1.0            | /    | . - .                                    |
| 0.20 – 0.40  | 1.0 – 3.0        | /    | ; .                                      |
| 0.40 – 0.60  | 3.0 – 5.0        | /    | . $\varepsilon$ ; .                      |
| 0.60 – 0.80  | 5.0 – 7.0        |      | . , .                                    |
| 0.80 – 0.95  | 7.0 – 10.0       | / КИ | $K_{\text{eff}}$ $R$ . . $\varepsilon$ . |
| 0.95 – 1.00  | > 10.0           | /    | . . - .                                  |

### 16.3 $L_c$

$L_c$  :

1. : , , . КИ YUCT (, , ).
2. : , ,  $L_c$ . , (, , ).
3. :  $L_c$  КИ , .
4. :  $L_c$  , , (, , , ).

$L_c$  YUCT, .

17 :

YUCT ( $K_{\text{eff}} \gg 1$ ). ; , , , , , . YUCT, ,  $K_{\text{eff}}$  .

17.1 (ECS):

(ECS) , , [45].  
ECS YUCT:

- $(\alpha)$ : , ECS. [46].
- $(\lambda)$ : ECS. ( $\lambda > 1$ ) , [45].

YUCT

ECS  $K_{\text{eff}}$ . , ECS  $(\kappa)$  ( $R$ ),  $K_{\text{eff}}$ . , , ECS ( $\epsilon$ ), ,  $K_{\text{eff}} \epsilon$ .  
[47].

17.2 :

[48]. , , , 100 [49]. ; , .

- : , . (de novo) , [50, 51].
- : , [52].

YUCT

,  $K_{\text{eff}}$ . ( $\tau_{\text{latency}}$ ) YPSDC. , ( $D_2$ ). , , ( $\epsilon$ ) ( $R$ ) . , ,  
[53, 54].

17.3 :

, , , (CSF) [55]. , , , КБ CSF.

YUCT

( $R$ ) . ( $\epsilon$ ), , ( $\epsilon$ , -) - . ( $\epsilon$ , -) , КБ  $R$  , [56].

17.4 : К Б

, , , , [57, 58].

- : , . , [59].
- : , , , , [60, 61].
- : , ( $\epsilon$ , -) [58].

YUCT

( $R$ ) YUCT. , , , КБ, . , .  $K_{\text{eff}}$ , , , ( $\epsilon$ , -),  
 $K_{\text{eff}}$  , ,  $\epsilon$ .

17.5 :

. ( ), , : , [62]. - (DTI) [63].

YUCT

$(D_2)$  . К НБ ,  $(\kappa)$  . , , .  $K_{\text{eff}}$  . (, - ) , .

17.6 : YUCT

, YUCT , . YUCT . .

5: YUCT .

| YUCT                 |                 |     |                    |
|----------------------|-----------------|-----|--------------------|
| $K_{\text{eff}}$ ( ) | (ECS),          | ,   | ECS ( $\alpha$ ),  |
| $R$ ( )              | , -             | ,   | ,                  |
| $\varepsilon$ ( )    | , ,             | , , | ECS ( $\lambda$ ), |
| $(D)$                |                 |     | DTI (FA),          |
|                      | , Glia limitans |     | CSF-               |

YUCT , , .

18 : VR,

YUCT,  $(\alpha_D, \beta_I, \gamma_R, K_{\text{eff}} \varepsilon)$ , , , , . , .

18.1

К НБ. YUCT , ,  $(D_2)$ ,  $(D_1)$   $(D_3)$ . , entry КНБ,  $\varepsilon_{\text{red}}$ .

18.1.1

(, - ), . VR/AR-, , :

- : , К -НБ  $(\varepsilon \rightarrow 0)$  , .
- : VR ,  $(R)$  , .
- КНБ :  $(K_{\text{eff}} \gg 1)$ , VR , , .

18.1.2

.  $\alpha_D, \beta_I \gamma_R$  , , VR- ,  $\varepsilon K_{\text{eff}}$ . , , , .

18.2 ,

$\{K_{\text{eff}}, R, \varepsilon\}$  .

### 18.2.1

- ,  $K_{\text{eff}} \varepsilon$  (, -),  $(\text{КНБ})$ , :
- :  $K_{\text{eff}} \varepsilon \cdot \text{VR-}$ , ,  $K_{\text{eff}}$  , .
  - : ,  $-\varepsilon \cdot \text{VR}$  ,  $-K_{\text{eff}} \text{КНБ}$ ,  $\gamma_R (-/)$   $\varepsilon$  ( ).
  - :  $\text{КНБ}$ , ,  $\text{КНБ} (L_c \approx 0.8 - 0.95)$ , .

### 18.3 :

- . YUCT ,  $\Sigma$ : YUCT [64].  
,  $\Sigma$  , , . :
- , : ; , .
  - YUCT: ,  $K_{\text{eff}}$  . , : , , .
  - :  $\text{КНБ}$   $\alpha_D, \beta_I, \gamma_R K_{\text{eff}}$  .
  - : , , .
- ;  $(\text{КНБ})$ , (ICTO) , UCD., YUCT,  $\Sigma$  .

### 18.4

YUCT  $\text{КНБ}$  . , , , . , , , .

## 19 : YUCT

YUCT , , :

- , , UCD.
- .
- , 19- .
- ,  $K_{\text{eff}}$  ( T).

### 19.1 YUCT:

#### 19.1.1 : D-I-R

YUCT , :

$$= D \oplus I \oplus R$$

:

- $D()$ : ,  $(, ,)$   $\text{КНБ}$  .
- $I()$ : ,  $(, ,)$   $\text{КНБ}$ .

- $R()$ :  $(, , )$  КИ, .

:

- :  $D() + I() + R() = .$  КИ  $D$ , КИ  $I$   $R$ .
- :  $D()$  ,  $I()$  ,  $R()$  .  $I$   $D$ ,  $R$ .
- :  $D()$  ,  $I()$  ,  $R()$  .

19.1.2 :

YUCT D-I-R:

$$= (D, I, R)$$

:

1. :  $D()$  .
2. :  $I()$  .
3. :  $R()$  .

$$\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta} : \varepsilon, , , K_{\text{eff}} \rightarrow \infty.$$

19.1.3 :

YUCT , :

$$, K_{\text{eff}}$$

$K_{\text{eff}}$  , . :

- $( / D)$  .
- $( I)$  .
- $/ ( R)$  .

; , ,  $K_{\text{eff}}(, , , )$ , , .

19.1.4

:

$$\frac{d(K_{\text{eff}}^{\text{humanity}})}{dt} > 0 \quad ( )$$

$(K_{\text{eff}} \approx 1.01)$   $(K_{\text{eff}} > 1.3)$  . Т , , (КИ) 20492055 ,  $K_{\text{eff}}$  , .  
YUCT : ,  $K_{\text{eff}}$  , :

- $(K_{\text{eff}} > 1.01)$  ( Т,  $K_{\text{crit}} \approx 150$ ).
- $(K_{\text{eff}} > 1.1)$  .
- $(K_{\text{eff}} > 1.5)$  .

; .

### 19.1.5

YUCT , :

- :  $D$  .
- :  $I$  .
- :  $R$  - .
- :  $I$  .
- :  $(R)$  КНБ , .

### 19.1.6 YUCT

YUCT , , , .  $D$ ,  $I$ ,  $R$ ,  $K_{\text{eff}}$  , .  
, YUCT , , : . , YUCT , 50 .  
/ , , .  $K_{\text{eff}}$  , .  
YUCT КНБ , , .

## 19.2 ,

, , , , . (UCD, H) D+IqR ,  $(K_{\text{eff}} \approx 8.5)$ . : , . UCD  
, . , , , , .

YUCT , . , , .

## 20

( , , ..), . YUCT , - . ,  $\varepsilon$ .

### 20.1 :

, , . YUCT ( ). ; .

### 20.2

, YUCT,  $(\varepsilon)$ , . ,  $(D_3)$ ,  $(R)$  . , , , , [66]. , .

- :  $\beta_I = H_{\text{int}}/H_{\text{ext}}$  . КНБ  $(\kappa)$ ,  $(H_{\text{ext}})$ , , entry. , .
- : , ,  $K_{\text{eff}}$ . , , , [68].
- : , (DMN),  $D_{\text{meta}}$  [67]. , , . КНБ , .



## 20.3

, .  $H_{\text{ext}}$  .

- $\beta_I$ : ,  $H_{\text{ext}}$ ,  $\beta_I$ . . . .
- $K_{\text{eff}}$ : , (  $D_2$  ) , . [69]. (  $R$  ) , .
- DMN : , DMN [70]. , , , .

## 20.4 YUCT

YUCT , : ,  $-K_{\text{eff}}$  , . ,  $-K_{\text{eff}}$  , . ; . , , .

## 20.5

, , .  $A \propto \varepsilon$ :

$$A \propto \varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta} \quad (2)$$

$\beta = 2/3$   $\kappa_c = 1/3$ .  $K_{\text{eff}}$  ( , ) , , , .

## 20.6

$\{K_{\text{eff}}, R, \varepsilon\}$ . :

- $K_{\text{eff}}$ : , . , , .
- $\varepsilon$ : , .
- :  $R$  , .

, , , . , , .

## 20.7

, .  $K_{\text{eff}}$  .  $K_{\text{crit}}$ , . :  $K_{\text{eff}}$  ( , ) ,  $K_{\text{eff}}$  .

## 20.8

:

1. : КНБ . , .
  2. : , YUCT, , . , , .
  3. : YUCT ,  $K_{\text{eff}}$  . , , , .
- YUCT, , , .

## 21

( , , . , . )

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